Tangible Interactions on a Flat Panel Display Using Actuated Paper Sheets

Kota Amano
The University of Tokyo
7-3-1 Hongo, Bukyo-ku
Tokyo, 113-8656, JAPAN
amano@aml.t.u-tokyo.ac.jp

Akio Yamamoto
The University of Tokyo
7-3-1 Hongo, Bukyo-ku
Tokyo, 113-8656, JAPAN
akio@aml.t.u-tokyo.ac.jp

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ITS'12, November 11–14, 2012, Cambridge, Massachusetts, USA.
ACM 978-1-4503-1209-7/12/11.

Abstract
This paper describes active tangible interactions on a flat panel display using plain paper sheets as tangible media. A prototype system employs transparent planar electrostatic actuators that cover the surface of a flat-panel display to realize co-located tangible and visual interactions. In a demonstration program, users can interact with animated computer graphics through plain paper sheets actuated by the electrostatic actuators.

Author Keywords
Interactive surface; tangible tabletop interface; transparent electrostatic actuator; paper sheet

ACM Classification Keywords
H.5.2 [Information interfaces and presentation (e.g., HCI)]: User Interfaces

General Terms
Design

Introduction
Tangible tabletop interfaces realize intuitive human-computer interaction using real tangible objects placed on computer screens as communication media [1,2]. In typical tangible interface systems, a user communicates with a computer by handling passive tangible objects, through which uni-directional interaction is realized.

The user experience would be enhanced considerably if the tangible objects can actively respond to the users' operations. Such active tangibles can be realized by using screen-embedded actuators [3,4]. As embedded actuators, most of the reported systems employed electro-magnetic motors. Although the use of electro-magnetic motors can provide high actuation power and high motion controllability, the resultant systems tend to have heavy and bulky structures due to the heavy use of ferromagnetic materials in those actuators. Another solution is to use self-moving tangibles that have built-in motors [5]. This allows easier integration of actuators but requires large and complicated tangible objects. For better integration of actuators to computer screens, another actuator solution would be preferred.

One promising alternative is electrostatic surface actuators [6,7]. Electrostatic surface actuators are characterized by its thin and light-weight structure, possible optical transparency, and actuation capability for everyday materials, such as plain paper sheets. In [8], a simple tangible interaction was demonstrated using an electrostatic actuator. The work emphasized its compact structure; the actuator is simply put on a flat panel display, which is not possible with typical electromagnetic motors. Although the work successfully showed the potential of the electrostatic actuators for active tangible interfaces, its demonstration was limited due to the opaqueness of the actuator used. The opaque actuator impeded visual rendering in the actuation area. To avoid this problem, the actuator covered only a limited area of the screen, and a special interaction scenario, *interactive shadow play*, was tailored, in which tangible and visual interactions took place at different locations.

This work tackles on this problem and realizes co-located tangible and visual interactions. Transparent electrostatic actuators are manufactured using indium-tin-oxide (ITO), a transparent conductive material. The use of transparent actuators allows to use a whole display as the interaction area, as well as to render computer graphics directly below the actuator to realize tangible action and visual rendering at the same place. This paper reports on the system structure of the prototyped tangible interface system and a demonstration program that realizes bi-directional user-computer interaction. It demonstrates an intuitive interaction with an animated character through a plain paper sheet as a tangible. The paper also mentions the potential simple force feedback in the prototype.

Transparent planar electrostatic actuator

Several actuation principles have been proposed for electrostatic surface actuators and this work utilizes induction actuation principle, which is realized by multi-phase electrode structure [7]. This work fabricated a three-phase electrode pattern using ITO on a transparent sheet of polyethylene terephthalate (PET) as shown in Figure 1. The three-phase electrodes and two bus lines for voltage feeding were made by ITO in the same layer, but the third bus line needs to be made in a different layer and was made by conducting silver paste, which is not transparent. The whole surface of the electrodes was covered with another PET sheet for insulation.

Actuation in one degree-of-freedom (DOF) is realized on the three-phase electrode area, whose size is 200 mm long and 100 mm wide. For induction actuation of a paper sheet, three-phase sinusoidal signals are applied to the three-phase electrodes [7]. The voltage



Appearance of the transparent actuator.

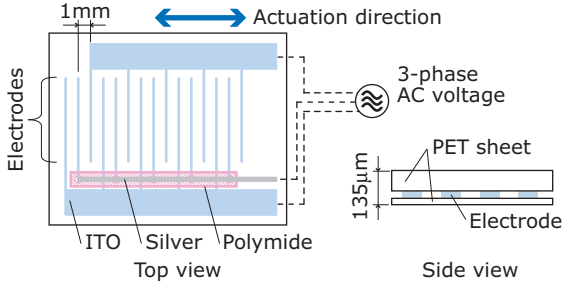


Figure 1: Structure of transparent electrostatic actuator.

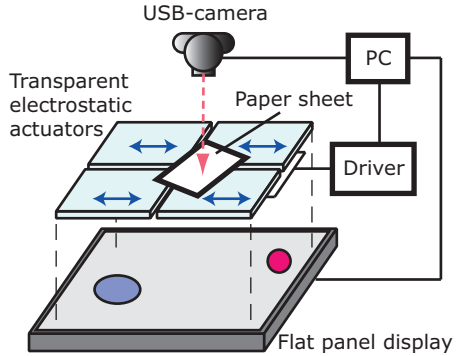


Figure 2: Setup overview of the interaction system.

amplitude and frequency in this work range between 600 V to 800 V and 100 Hz to 1 kHz, respectively. The motion of the actuated sheet is asynchronous and requires feedback control. This work utilizes a camera as a position sensor. To cope with the limited operation speed of the camera (30 frames per second), the actuator was intermittently activated.

Active tangible interface system

System overview

Figure 2 shows the schematic of the prototype system. Four electrostatic actuators are placed on a flat panel display to cover a most part of the surface of 24.1 inch wide liquid crystal display. All the actuators have 1-DOF motion capability in lateral direction and thus the motion of tangible objects is restricted in the lateral 1-DOF, in this particular work. On the surface of the actuators, glass beads with 100 μm diameter are scattered to keep spacing with the actuated objects and to reduce the surface friction. The resultant friction coefficient was about 0.07. Tangible objects are sheets of plain paper, on which illustrations are printed by a laser printer and are cut into corresponding shapes. For actuation of a tangible paper sheet, a computer program sends actuation commands to the driver electronics. In this particular system, the driver electronics activates all the four actuators at the same time in the same actuation condition.

Co-located tangible interaction using plain paper

A pilot application program was implemented to demonstrate active tangible interactions. The program draws an animated cat, which wanders around in the whole screen. At the same time, the program detects the position of a paper sheet by using the overhead camera. The animated cat is programmed to rush to the detected paper sheet. As any collisions occur between the paper sheet and the animated cat, the program issues an actuation command so that the paper sheet will be moved in an appropriate lateral direction.

With the implemented program, a user can play with the animated cat through the paper sheet. User experience may differ depending on the design of the paper sheet. For example, if the paper sheet is a ball, a user can play catch with the cat; as the user throws the ball to the cat, the cat hits it back to the user (see

Note:
As reported in [7], performance of this actuation principle is sensitive to humidity. The experiments of this work were performed at relative humidity of 45-60%. Driving signal frequency was adjusted depending on the condition.

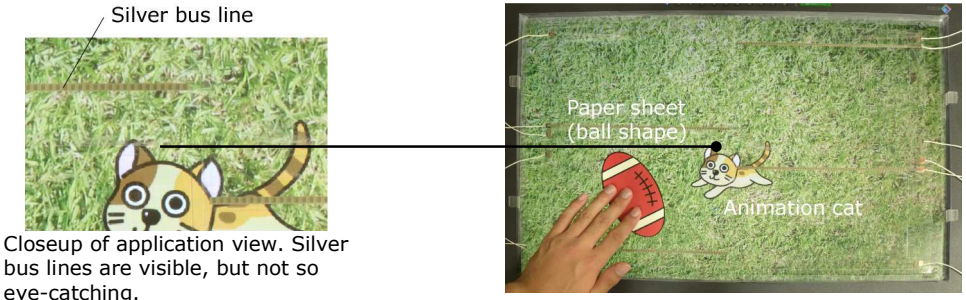
Figure 3). Or, if the sheet is designed as a foxtail, a user can pat the cat as shown in Figure 4.

Force perception

The electrostatic force generated on the paper sheet is just enough to move the paper sheet and cannot be clearly perceived by a user in direct touch as in Figure 3. However, with a long sheet such as foxtail as in Figure 4, the force can be amplified by a lever mechanism, and the user could feel as if the animated cat was pushing back through the sheet.

Conclusion

This paper demonstrated that transparent electrostatic actuators and plain paper sheets as tangible media can realize co-located tangible and visual interactions with a simple system structure. Although this particular work dealt with a simple interaction in 1-DOF motion, future studies will address more sophisticated interactions by multi-DOF motion and multi-objects actuation.



Closeup of application view. Silver bus lines are visible, but not so eye-catching.

Figure 3: Example of the active tangible interaction. The user is playing catch with animation cat.

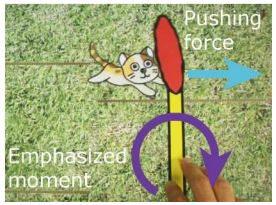


Figure 4: Force amplification for force perception.

Acknowledgement

This work was supported by Funding Program for Next Generation World-Leading Researchers (LR013) from JSPS, Japan.

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